mortgages would be extremely vulnerable to a complete loss.

The story, however, does not end there. The BBB tranches were difficult to sell. Wall Street alchemists came up with a solution that magically transformed the BBB tranches into AAA. They created a new securitization called a *collateralized debt obligation* (CDO) that consisted entirely of the BBB tranches of many mortgage bonds.[7] CDOs also employed a tranche structure. Typically 75 percent to 80 percent of a CDO was rated AAA, even though it consisted of 100 percent BBB tranches.

Although the CDO tranche structure was similar to that employed by subprime mortgage bonds consisting of individual mortgages, there was an important difference. In a pool of mortgages, there was at least some reason to assume there would be limited correlation between individual mortgages. Different individuals would not necessarily come under financial stress at the same time, and different geographic areas could witness divergent economic conditions. In contrast, all the individual elements of the CDOs were clones—they all represented the lowest tier of a pool of subprime mortgages. If economic conditions were sufficiently unfavorable for the BBB tranche of one mortgage bond pool to be wiped out, the odds were very high that BBB tranches in other pools would also be wiped out, or at least severely impaired.[8] The AAA tranche needed approximately a 20 to 25 percent loss to begin being impaired, which sounds like a safe number, until one considers that all the holdings are highly correlated. The BBB tranches were like a group of people in close quarters contaminated by a highly contagious flu. If one person is infected, the odds that many would be infected would increase dramatically. In this context, the 20 percent cushion of the AAA class sounds more like a tissue paper layer.

How could bonds consisting of only BBB tranches be rated AAA? There are three interconnected explanations.

1. Pricing models used historical data on mortgage defaults. Historical mortgages in which the lender actually cared whether repayments were made and required down payments and verification bore no resemblance to the more recently minted no-down-payment, no-verification loans. Therefore, historical mortgage default data would grossly understate the risk of more recent mortgages defaulting.